

The impact of different antimicrobial exposures on the gut microbiome in the ARMORD observational study

Reviewed Preprint

v1 • July 16, 2024

Not revised

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Abstract

Background

Better metrics to compare the impact of different antimicrobials on the gut microbiome would aid efforts to control antimicrobial resistance (AMR).

Methods

The Antibiotic Resistance in the Microbiome – Oxford (ARMORD) study recruited inpatients, outpatients and healthy volunteers in Oxfordshire, UK, who provided stool samples for metagenomic sequencing. Data on previous antimicrobial use and potential confounders were recorded. Exposures to each antimicrobial were considered as factors in a multivariable linear regression, also adjusted for demographics, with separate analyses for those contributing samples cross-sectionally or longitudinally. Outcomes were Shannon diversity and relative abundance of specific bacterial taxa (*Enterobacteriaceae*, *Enterococcus*, and major anaerobic groups) and antimicrobial resistance genes (targeting beta-lactams, tetracyclines, aminoglycosides, macrolides, and glycopeptides).

Results

225 adults were included in the cross-sectional analysis, and a subset of 79 patients undergoing haematopoietic stem cell transplant provided serial samples for longitudinal analysis. Results were largely consistent between the two sampling frames. Recent use of piperacillin-tazobactam, meropenem, intravenous co-amoxiclav and clindamycin were associated with large reductions in microbiome diversity and reduced abundance of anaerobes. Exposure to piperacillin-tazobactam and meropenem were associated with a decreased abundance of *Enterobacteriaceae*, and an increased abundance of *Enterococcus* and major AMR genes, but there was no evidence that these antibiotics had a greater impact

on microbiome diversity than iv co-amoxiclav or oral clindamycin. In contrast, co-trimoxazole, doxycycline, antifungals and antivirals had less impact on microbiome diversity and selection of AMR genes.

Conclusion

Simultaneous estimation of the impact of over 20 antimicrobials on the gut microbiome and AMR gene abundance highlighted important differences between individual drugs. Some drugs in the WHO Access group (co-amoxiclav, clindamycin) had similar magnitude impact on microbiome diversity to those in the Watch group (meropenem, piperacillin-tazobactam) with potential implications for acquisition of resistant organisms. Metagenomic sequencing can be used to compare the impact of different antimicrobial agents and treatment strategies on the commensal flora.

eLife assessment

The work provides a **valuable** assessment of how antibiotics impact the human gut microbiota in diverse observational cohorts. Although the data presented are **solid**, some of the assumptions underlying their models may have affected the interpretation of their findings. The study is relevant for researchers and clinicians interested in antimicrobial resistance.

<https://doi.org/10.7554/eLife.97751.1.sa2>

Introduction

Effective antimicrobial stewardship is necessary to limit the emergence and spread of antimicrobial resistance (AMR),¹ and this includes the preferential use of agents least likely to select for drug resistant pathogens among the commensal flora. This typically consists of avoiding “broad” spectrum antimicrobials, as well as those to which resistance is currently rare, and these principles underlie the World Health Organisation AWaRe classification (Access, Watch and Reserve).² AWaRe is reflected in the current UK National Health Service standard contract, which requires hospitals to increase the proportion of “Access” antibiotics used, replacing previous requirements to reduce overall antibiotic use.³ However, these classifications are largely based on activity against pathogens rather than direct measures of AMR gene selection or microbiome disruption, and stewardship could be improved if such measures were available. For example, two antibiotics being considered for use may have similar spectra against a target pathogen, but very different impacts on AMR selection, either because the amounts reaching the commensal flora differ, or because they have differing spectra against non-pathogenic commensals that protect against colonisation with drug-resistant pathogens. Metagenomic sequencing provides a direct measure of AMR genes and microbiome composition in a sample, and its increasing availability in recent years is starting to allow the fuller impacts of antimicrobials to be measured.

The large intestine contains the vast majority of human commensal bacteria and is the primary reservoir for several clinically important commensal pathogens, particularly the *Enterobacteriaceae* (including *Escherichia coli* and *Klebsiella pneumoniae*) and *Enterococcus faecium*. Increasing multi-drug resistance in these organisms represents a major global public health challenge.^{4,5} Existing evidence linking antibiotic exposure to individual-level selection

for AMR in the gut flora comes predominantly from small, healthy volunteer studies, which have shown that antibiotics can cause rapid microbiome disruption, but provide limited comparative data between antibiotics. They also may also have limited applicability to real patients, who often have recent exposure to several different agents and are at high risk of colonisation with drug resistant organisms. Randomised trials are a robust method to assess the impacts of different treatment approaches, but few have reported microbiome outcomes.^{6,7,8,9} Another approach is to exploit variation in routine use of antibiotics in groups of patients at high risk of AMR to understand the nature and extent of differences between agents which cannot easily be achieved in other designs. Here we report results from a prospective observational study assessing the impact of antimicrobial use on the gut microbiome and selection of AMR genes. This study included two different sampling frames to produce independent and complimentary estimates, one cross-sectional, analysing a single stool sample from each participant, and one longitudinal, analysing changes in serial samples taken from the same participant admitted for haematopoietic stem cell transplant to enrich for broad spectrum antimicrobial exposure.

Methods

Study design and participants

The Antibiotic Resistance in the Microbiome – Oxford (ARMORD) study was an observational study that recruited healthy individuals living in Oxfordshire, and patients at the Oxford University Hospitals NHS Foundation Trust (OUH). The study was coordinated by the Nuffield Department of Medicine, University of Oxford, and was approved by the East Midlands-Leicester Central Research Ethics Committee (15/EM/0270).

The study involved two sampling strategies:

1. *Cross-sectional sampling.* Participants provided a single stool sample, and measures of the microbiome and AMR gene abundance were related to exposures recorded at the time of sampling.
2. *Longitudinal sampling.* Participants provided serial stool samples, and changes in the microbiome and AMR gene abundance between serial samples from the same individual were related to exposures between samples. Longitudinal sampling was only performed in patients admitted to the OUH haematology ward for haematopoietic cell transplant (HCT). The initial sample collected from these patients was also used in the cross-sectional analysis.

Participants were eligible if they were ≥ 18 years old, had no stoma or active inflammatory bowel disease, and were able to give informed consent and provide a history of recent antibiotic use. Inpatients and outpatients at OUH were approached by a member of the study team during routine care, and healthy individuals responded to articles in local media. After providing written informed consent, participants were interviewed and their medical notes were reviewed to collect information about antimicrobial exposures in the past year, recent travel, diet, alcohol and tobacco use, animal exposures, healthcare exposure and use of specific drugs (case report form in Appendix p8). Electronic patient records available at OUH included i) inpatient, emergency department and outpatient attendances, ii) inpatient and discharge antimicrobial prescriptions iii) microbiology, haematology and biochemistry results, iv) inpatient clinical observations (used to calculate the National Early Warning Score 2 [NEWS2] summary score of physiological abnormality),¹⁰ and v) discharge coding (including Charlson co-morbidity index).

Sample collection

In the cross-sectional stratum, participants were asked to provide the first stool sample passed after recruitment. This was stored at ambient temperature for a maximum of 24 hours before being frozen at -80°C . In the longitudinal stratum, participants were asked to provide a stool sample every other day until discharge. These were stored at $4-8^{\circ}\text{C}$ for up to 72 hours before being frozen at -80°C .

DNA extraction, sequencing and bioinformatic analysis

DNA extraction was performed by bead beating in Lysing Matrix E tubes (MP Biomedicals) followed by QIAGEN Fast DNA Stool MiniKit (QIAGEN) (details in Appendix p1). Samples were sequenced in batches of 56-114 at the Oxford Centre for Genomics, and had automated normalisation and library preparation using either NEBNext Ultra or NEBNext Ultra II FS kits. All samples from the same batch were pooled and sequenced across 2-8 Illumina HiSeq4000 lanes using 150bp paired-end sequencing. Following human read removal all samples were subsampled to a depth of 3.5 million paired reads, and samples with fewer reads were excluded. Taxonomic classification was performed with MetaPhlAn2 (for diversity indices) and Kraken2 (for abundance of specific taxa). AMR gene detection in metagenomic sequence data was performed with the ARIBA software package, using the CARD database and ontology (v3.0.2) (details in Appendix p2).

Outcomes

Sequence data was used to derive three types of outcome:

1. *Shannon diversity index* - a single metric of diversity for each sample (calculated as the sum of $p \cdot \ln(p)$ for all species, where p is proportional abundance). This index incorporates abundance and the number of species detected (richness).
2. *Log relative abundance of specific bacterial taxa*. The taxa of interest were two major groups of opportunistic pathogens (family *Enterobacteriaceae* and genus *Enterococcus*), and three major groups of anaerobes (phyla Bacteroidetes and Actinobacteria, and class Clostridia). Together these non-overlapping taxa accounted for 89% of the total abundance of organisms across all samples. If a particular taxon was not detected, its relative abundance was imputed as 10^{-6} (the approximate lower limit of detection).
3. *Log relative abundance of specific classes of AMR genes*. Five classes of clinically important resistance mechanisms were of interest: Clinically significant beta-lactamases (CTX-M, OXA, TEM, SHV), tetracycline ribosomal protection proteins, aminoglycoside transferases (AAC, ANT, APH), macrolide/clindamycin resistance genes (*erm* and *mef*), and the *vanA* vancomycin resistance gene. If a particular gene was not detected its relative abundance was imputed as 10^{-5} (the approximate lower limit of detection). Further details are in Appendix p3.

Statistical analysis

Cross-sectional sampling frame

Multivariable linear regression was used to estimate the effects of specific antimicrobial exposures on the outcomes above. Covariates were: age, sex, participant category (healthy, general medical, autologous stem cell transplant, allogeneic stem cell transplant), days of chemotherapy received (0 for non-HCT participants, truncated at 14), maximum Charlson comorbidity score in the year before sampling (identified from electronic health records), and the following physiological abnormalities in the fortnight prior to sample collection; maximum NEWS2 score, C-reactive protein (CRP) >50 mg/dL, white cell count (WCC) $>11 \times 10^9/\text{L}$, and WCC $<0.5 \times 10^9/\text{L}$. Healthy

volunteers lived in the OUH catchment area but most had no previous activity at OUH, and in this case normal values were imputed (i.e. Charlson Index 0, NEWS2 score 0, CRP <50 mg/dL, WCC <11 and $>0.5 \times 10^9/L$). Other covariates such as diet and travel were not included in the final model, as they were not significantly associated with Shannon diversity in multivariable analysis and their inclusion did not materially affect other estimates. We did not make formal adjustment for multiple testing, as many of the antibiotic exposures were correlated, but interpreted level of evidence supporting findings in the context of the number of comparisons performed.

All antimicrobial exposures observed in >5 participants were included in the model. Individual agents were categorised separately if given by different administration routes (e.g. oral vancomycin and intravenous vancomycin), but route of administration was ignored in analyses of antimicrobial class (e.g. glycopeptides). In categorising antimicrobial class, beta-lactams were divided into 'narrow spectrum' (defined as penicillin, amoxicillin, flucloxacillin and first generation cephalosporins) and 'broad spectrum' (all others). Exposure to each antimicrobial was included as a separate variable on a scale of 0 (no exposure) to 1 (full exposure). In order to reflect both recency and total duration of antimicrobial use, this exposure was modelled as the area under an exponential decay curve of the form $y = 2^{-x/\lambda}$, in which λ is the microbiome disruption half-life, and x is time before sample collection. A single value of λ was used for all analyses, chosen as the common value across all antimicrobial exposures with the lowest Akaike Information Criterion across 1 to 14 days in the cross-sectional model for microbial diversity (6 days; Supplementary Table 1). The disruption half-life of 6 days means that after 6 days of an antibiotic course a patient would have an exposure of 0.5 to that agent, and after 12 days they would have an exposure of 0.25. Details of the exposure calculation, including graphical depictions are in the Appendix (p3 and Supplementary Figures 1-2).

Longitudinal sampling frame

Serial samples collected from participants undergoing HCT were used for the longitudinal analysis. The unit of analysis was a pair of consecutive samples collected from the same individual. For patients with >2 samples, each consecutive pair was used (i.e. sample 2 in pair 1 was sample 1 in pair 2, and so on, so the total number of pairs per participant is one less than the number of samples). Only pairs collected within 2-30 days were used. A multivariable linear regression model was used that was analogous to the cross-sectional model described above, except that the outcome was the *change* between the first and second samples in a pair, rather than absolute values (i.e. change in Shannon diversity, or change in log relative abundance of bacterial taxa or AMR genes). Because pairs of samples from the same participant may not be independent of one another, robust standard errors were used to allow for possible clustering.

Covariates were: age, sex, type of transplant (allogeneic or autologous), days of chemotherapy received at collection of sample 1 (0 in non-HCT patients, and truncated at 14), number of days between sample 1 and sample 2, change in NEWS2 score between sample 1 and sample 2, and the presence of the following new physiological abnormalities recorded after collection of sample 1 and before collection of sample 2; CRP >50 mg/dL, WCC $>11 \times 10^9/L$, WCC $<0.5 \times 10^9/L$. For each outcome, the value for the first sample in the pair was also included as a covariate (i.e. baseline diversity or log relative abundance of taxa/AMR genes).

Antimicrobial exposures were calculated for samples as in the cross-sectional analysis above, and the exposure for each pair was the difference between the first and second samples. This has the following implications: i) if a patient starts an antimicrobial after the first sample is collected then the exposure for that pair is the same as for sample 2, ii) if a patient is on long-term antimicrobial treatment then the exposure for a pair is zero (as one would not expect this to lead to a difference between samples), iii) if a patient stops antimicrobial treatment shortly after sample 1 then the

exposure to that agent will be negative (as one expects gut microbiome diversity to increase after stopping an antimicrobial). Truncating the small number of negative values at zero had little impact on results (data not shown).

Analysis was performed in R v4.2.3 with the ontologyIndex package (v2.4).

Results

Between July 2015 and November 2018, 225 participants were recruited and had at least one sequenced stool sample, all of whom were included in the cross-sectional analysis (**Table 1** and Supplementary Figure 3 [CONSORT diagram]). Thirty-three (15%) were healthy volunteers, 91 (40%) were general medical patients, and 101 (45%) were HCT patients. The healthy volunteers were on average younger, more likely to be female, and rarely had recent antibiotic exposure. In contrast, 184/192 (96%) of medical and haematology patients had received antibiotics in the past year, and 97 (51%) of these were receiving antibiotics at the time of sampling. Of those with antibiotic exposure in the past month, 99/148 (67%) had received >1 type of antimicrobial. Of 101 HCT participants in the cross-sectional analysis, 79 had >1 sequenced sample and so also contributed to the longitudinal analysis, and 173 sample pairs were included in this analysis.

Impact of antimicrobial exposures on gut microbiome diversity

A half-life of 6 days was identified as the best fit for the antimicrobial exposure in the cross-sectional model of microbiome diversity, and this value was used for all subsequent analyses (Supplementary table 1). The independent effects of antimicrobial exposure on gut microbiome diversity in the cross-sectional and longitudinal models are shown in **Figure 1**. The cross-sectional model (**Figure 1A**) provided more precise estimates of microbiome disruption than the longitudinal model, and four antimicrobial exposures were associated with a significant reduction in gut Shannon diversity: iv co-amoxiclav (-3.0 [95% Confidence Interval -4.7 to -1.4 ; $p = 0.0005$), iv piperacillin-tazobactam (-2.6 , [-3.4 to -1.8]; $p < 10^{-9}$), oral clindamycin (-2.2 [-3.5 to -0.8]; $p = 0.002$), and iv meropenem (-1.6 [-2.6 to -0.5]; $p = 0.003$). There was no evidence that the impact of iv co-amoxiclav was greater than iv meropenem ($p = 0.13$). In contrast, co-trimoxazole and doxycycline were associated with minimal reductions in gut microbiome diversity, as were azole antifungals and acyclovir (lower 95% CI above -1.0). In the longitudinal analysis (**Figure 1B**), only five exposures had data from more than 20 sample pairs, and these were consistent with the cross-sectional results, including large reductions in diversity associated with exposure to piperacillin-tazobactam (-3.9 [-5.0 to -2.9]; $p < 10^{-10}$) and meropenem (-3.0 [-4.1 to -1.9]; $p < 10^{-6}$) (data on iv co-amoxiclav and oral clindamycin insufficient for comparison). In the longitudinal model, gentamicin was associated with marginally significant increased gut microbiome diversity ($+6.6$ [$+1.3$ to $+11.8$]; $p = 0.02$).

Analyses by antimicrobial class were also largely consistent between cross-sectional and longitudinal analyses (Supplementary Figure 4). In both analyses ‘narrow’ spectrum beta-lactams were associated with substantially lower microbiome disruption than ‘broad’ spectrum beta-lactams. In the cross-sectional analysis, several other classes of antibiotics also had little to no impact on the microbiome diversity; antifolates, macrolides, aminoglycosides and tetracyclines (lower 95% CI above -1.0).

	Healthy volunteers (n=33)	General medical patients (n=91)	HCT patients (n=101)	All participants (n=225)
Age, years (median, IQR)	37 (31-49)	76 (67-83)	58 (50-66)	64 (50-73)
Sex (n, %)				
Male	7 (21%)	53 (58%)	60 (59%)	120 (53%)
Female	26 (79%)	38 (42%)	41 (41%)	105 (47%)
Recent antibiotic use (n, %)				
Receiving antibiotics at time of sampling	0 (0%)	55 (60%)	42 (42%)	97 (43%)
Use in past month (but not at time of sampling)	3 (9%)	26 (29%)	22 (22%)	51 (24%)
Use in past year (but not in past month)	4 (12%)	6 (7%)	33 (33%)	43 (19%)
No antibiotics in past year	26 (79%)	4 (4%)	4 (4%)	34 (15%)
Max Charlson index in past year (median, IQR)*	0 (0-0)	4 (0-13)	0 (0-8)	0 (0-8)
Maximum values in past 14 days (median, IQR)				
NEWS2*	0 (0-0)	5 (2-8)	3 (2-4)	3 (1-5)
C-reactive protein†	0.2 (0.2-0.2)	63 (22-163)	10 (3-69)	21 (2-81)
White cell count‡	7.5 (7.5-7.5)	11.5 (8.4-14.4)	7.6 (5.8-10.9)	8.4 (7.4-12.4)
Days of chemotherapy at time of sampling (median, IQR)	0 (0-0)	0 (0-0)	3.0 (1.4-7.2)	0 (0-2.8)

* National Early Warning Score 2. Imputed as 0 if no observations recorded, see Methods.

† Imputed as 0.2 if no result recorded

‡ Imputed as 7.5 if no result recorded

Table 1

Characteristics of participants in cross-sectional analysis

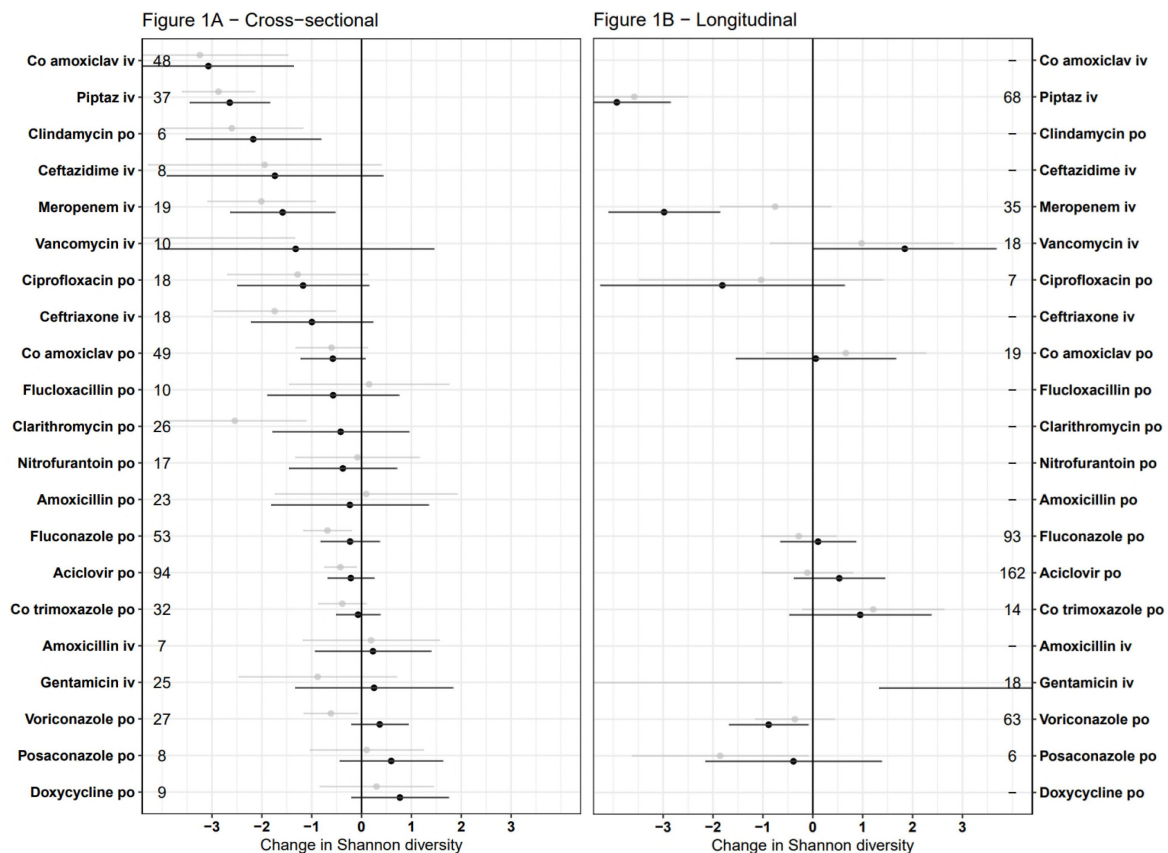


Figure 1 -

Independent effect of specific antimicrobial exposures on Shannon diversity in A) cross-sectional, and B) longitudinal analyses.

Multivariable estimates are in black, univariable (unadjusted) estimates in grey. Error bars represent 95% confidence intervals. Non-antimicrobial covariates are not shown but were included in the model and can be found in supplementary data. Results are not plotted for 4 antimicrobials (n = 6-12) that had standard errors >3 and did not differ significantly from zero. Estimates represent the impact of prolonged use, when exposure = 1 (approximately 42 days, see Supplementary Figure 2).

Effects of antimicrobials on the abundance of specific taxa and AMR genes

Piperacillin-tazobactam, meropenem, and ciprofloxacin were associated with significant decreases in the relative abundance of *Enterobacteriaceae* in both models, as were ceftriaxone and ceftazidime in the cross-sectional model (>1,000 fold decreased relative abundance, $p < 0.01$, **Figure 2** & Supplementary Figure 5). There was no evidence of effects of iv co-amoxiclav ($p = 0.90$) or oral clindamycin ($p = 0.31$) on relative abundance of *Enterobacteriaceae*. Piperacillin-tazobactam and meropenem were also associated with large increases in the relative abundance of *Enterococcus* in both models (>100 fold increased relative abundance, $p < 0.01$), as were iv co-amoxiclav ($p = 0.04$), iv ceftriaxone ($p = 0.02$) and iv vancomycin ($p = 0.03$) in the cross-sectional model. Many antibiotics were associated with large reductions in the relative abundance of major anaerobe groups, particularly Actinobacteria (reductions in which were associated with exposure to ceftazidime, oral ciprofloxacin, oral clindamycin, iv co-amoxiclav, meropenem, piperacillin-tazobactam, and iv vancomycin). When analysed by antimicrobial class, broad spectrum beta-lactams were associated with reductions all anaerobe groups (Supplementary Figure 5).

Several antimicrobial exposures were also associated with increases in the relative abundance of AMR genes (**Figure 3**, Supplementary Figure 6). In the cross-sectional analysis, iv piperacillin-tazobactam and meropenem were both associated with increases in several AMR mechanisms, including aminoglycoside transferases and macrolide resistance genes (all $p < 0.02$) but there was no evidence that these antibiotics had any effect on overall relative abundance of beta-lactamases ($p \geq 0.14$), potentially reflecting lower abundance of relative species (see above) but greater AMR gene carriage in those surviving. In the cross-sectional model, exposure to several classes of antimicrobials were associated with an increase in the abundance of corresponding resistance genes, including glycopeptides, tetracyclines, macrolides and clindamycin (all $p \leq 0.02$). Broad-spectrum beta-lactam use was not associated with an increase in the abundance of beta-lactamases, but it was associated with an increase in the abundance of glycopeptide and aminoglycoside AMR genes (both $p \leq 0.0001$). Increased aminoglycoside AMR gene abundance was also associated with clindamycin and glycopeptide use (both $p \leq 0.02$), but not aminoglycoside use.

Discussion

Our findings demonstrate that simultaneous modelling of multiple antimicrobial exposures in a heterogeneous and heavily antibiotic exposed population can produce direct, quantitative comparisons of the impact of different agents on multiple aspects of the gut microbiome. Several broad-spectrum antibiotics were associated with large and rapid reductions in gut microbiome diversity, including decreased abundance of several major anaerobe groups, and increased abundance of *Enterococcus* species, along with glycopeptide and aminoglycoside resistance mechanisms often found in *Enterococcus faecium*.⁵ The plausibility of these results is reinforced by the consistency between the two independent analysis frameworks that were used; cross-sectional and longitudinal. Microbiome disruption was clearest with clindamycin and broad spectrum beta-lactams including iv co-amoxiclav, piperacillin-tazobactam, and meropenem, consistent with some previous microbiome studies of these drugs.^{11,12-14} This is also in keeping with estimates of risk of *Clostridioides difficile* infection (CDI) which is highest with clindamycin, fluoroquinolones, carbapenems and third-generation cephalosporins.^{15,16} The limited impact of doxycycline on diversity is also consistent with the lower risk of CDI observed with tetracyclines. The decreased relative abundance of *Enterobacteriaceae* seen in this study with piperacillin-tazobactam, ceftriaxone, meropenem, and ciprofloxacin correspond with reductions seen in culture based studies.¹⁷ The absence of any clear effect of beta-lactams on the

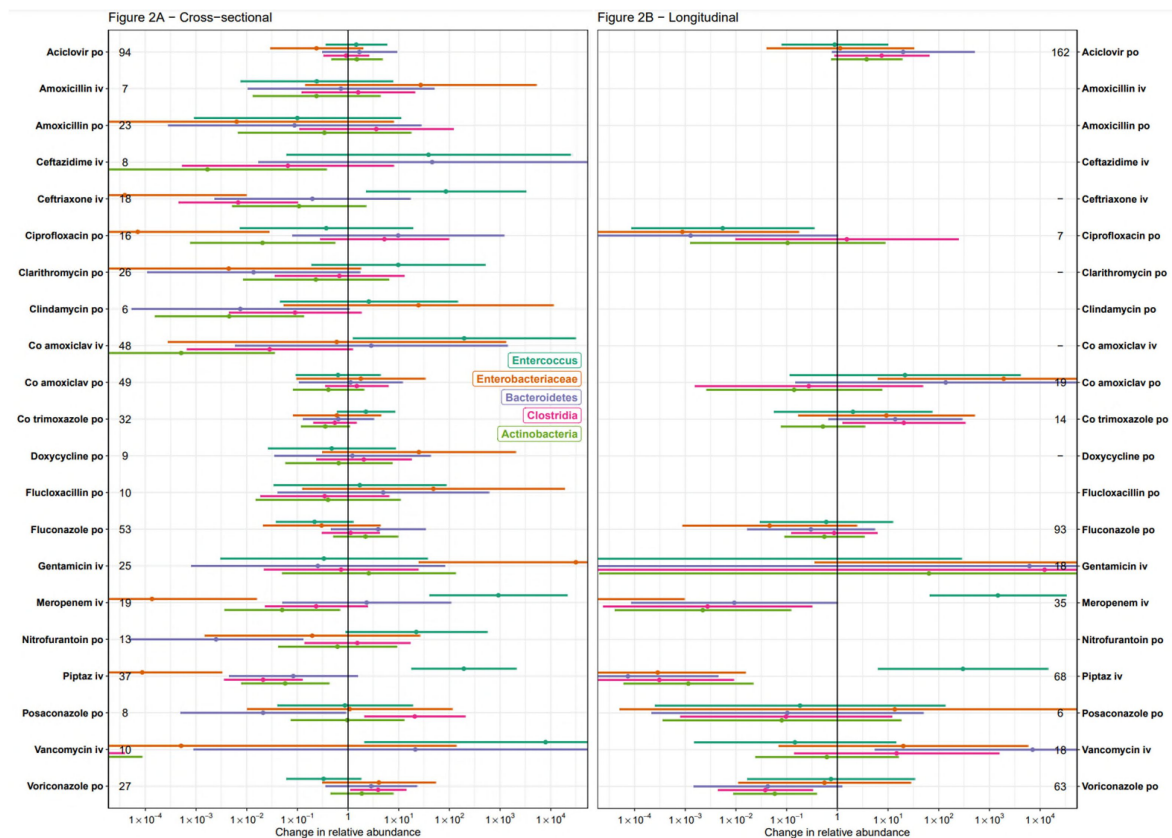


Figure 2 -

Independent effects of specific antimicrobial exposures on relative abundance of selected taxa in A) cross-sectional, and B) longitudinal multivariable analyses.

Error bars represent 95% confidence intervals. Non-antimicrobial covariates are not shown but were included in the model and can be found in supplementary data. Estimates represent the impact of prolonged use, when exposure ≈ 1 (approximately 42 days, see Supplementary Figure 2).

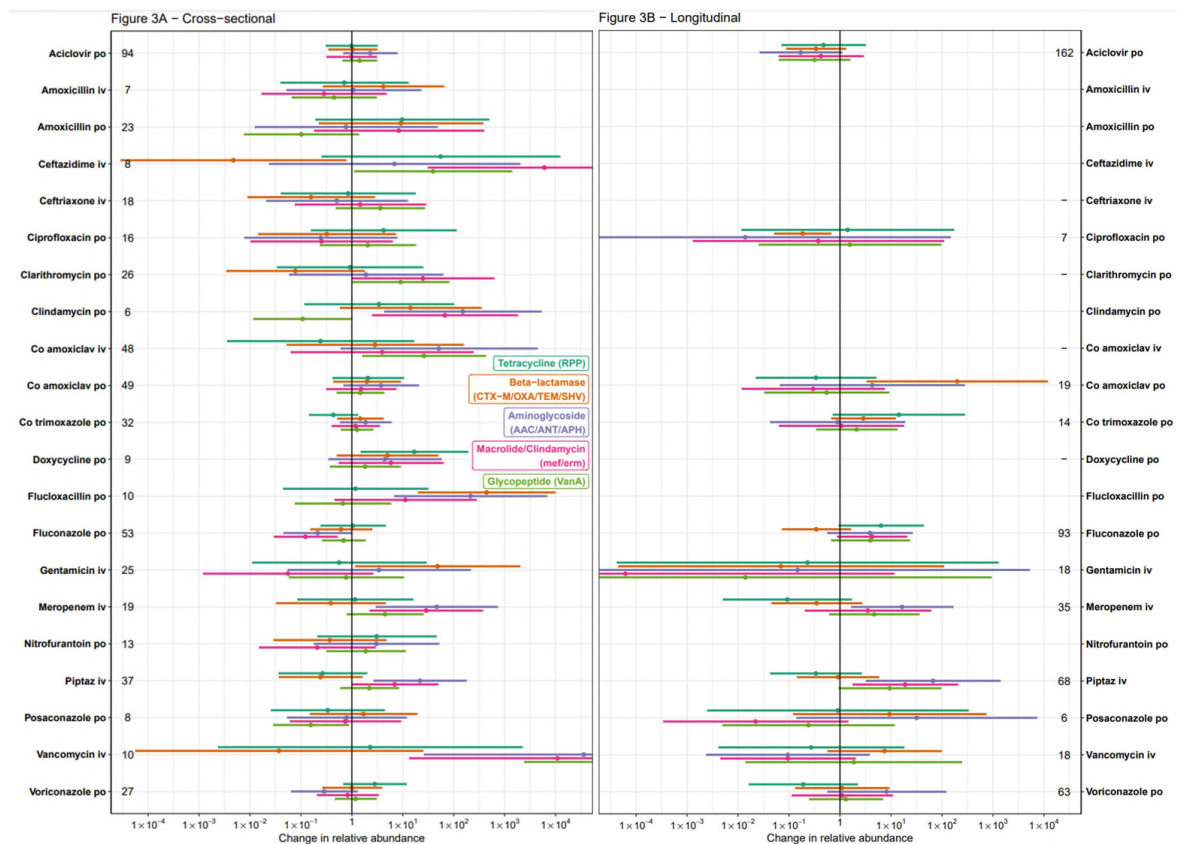


Figure 3 -

Independent effects of specific antimicrobial exposures on relative abundance of selected AMR genes in A) cross-sectional, and B) longitudinal multivariable analyses.

Error bars represent 95% confidence intervals. Non-antimicrobial covariates are not shown but were included in the model and can be found in supplementary data. Estimates represent the impact of prolonged use, when exposure ≈ 1 (approximately 42 days, see Supplementary Figure 2).

abundance of beta-lactamase gene abundance may be because there were too few patients in this study to create robust categories of beta-lactamase by resistance spectrum, so opposite impacts on different beta-lactamase genes would have been combined.

Observational studies of the gut microbiome allow large numbers of patients to be recruited much more easily than interventional studies such as clinical trials. However, using data from these to inform antibiotic usage is complicated by potential biases from confounding, and because of difficulties in accurately modelling multiple exposures in patients receiving many different antibiotics. In ARMORD, the apparent impact of vancomycin and clarithromycin on diversity was substantially reduced when adjusting for other antibiotic exposures (**Figure 1A**), in keeping with the fact that these drugs are typically given alongside broad-spectrum beta-lactams in the UK. Nevertheless, the complexity of antibiotic exposure captured in observational data also more closely reflects real life, whereas clinical trials generally manipulate one single antibiotic, and restrict background antibiotic exposure to produce a cleaner, but potentially less generalisable, comparison.

The large degree of inter-individual variation in the gut microbiome introduces an additional source of variation to cross-sectional studies compared to longitudinal sampling. Despite this, cross-sectional sampling in ARMORD generally gave more precise estimates than longitudinal sampling, despite using a similar number of samples. Recruiting any hospitalised patients for longitudinal sampling before they have received antibiotics is challenging, and this was true in ARMORD, even among patients admitted electively for HCT. In ARMORD the quasi-experimental approach of longitudinal sampling starting before antimicrobial exposure had no advantage over cross-sectional sampling, and complicated recruitment.

The ARMORD study has important limitations. Along with age and sex, several markers of acute illness and comorbidity were included in the model to adjust for potential confounding, but there may be residual and unmeasured confounding related to factors not adequately represented in the model. The microbiome disruption half-life used was 6 days, as this value best fitted the overall data, which implies that the majority of the disruption and recovery of the bowel flora diversity occurs rapidly after starting and stopping antimicrobials (Supplementary Figure 2). This is in keeping with interventional studies in volunteers,^{13,18} but is a simplification that does not account for some longer term impacts of antibiotic use that can be detected months after treatment.^{19,20} A larger study may have allowed derivation of exposure models that take this into account, or could test multiple different ways to represent the impact of antibiotic exposure on the gut microbiome, or that allow for different disruption dynamics for different antimicrobials. Also, because the study was observational, no data were available on drugs that are not commonly used at OUH, including imipenem, cefepime and aztreonam. Finally, although we estimate the independent effects of multiple exposures, the uncertainty is still large making it challenging to use these results to adequately inform clinical use, as this would require clear distinction between antibiotics that might be given for the same indication. For example, the cross-sectional data are consistent with an identical average reduction in diversity with piperacillin-tazobactam, meropenem, ceftazidime, ceftriaxone, intravenous co-amoxiclav, ciprofloxacin and clindamycin, but they are also consistent with important differences between these drugs. A larger study is required to identify or rule out such differences.

Overall, however, simultaneous estimation of the impact of over 20 antimicrobials on the gut microbiome and AMR gene abundance highlighted important differences between individual drugs, and that some drugs in the WHO Access group (co-amoxiclav, clindamycin) had similar magnitude impact on microbiome diversity to those in the Watch group (meropenem, piperacillin-tazobactam) with potential implications for acquisition of other resistant organisms. The consistency of the ARMORD results between sampling frames and with previous studies supports

the wider use of observational metagenomic studies to compare the impact of antimicrobials on the gut microbiome. Although some challenges remain, such as identifying an optimal measure of antimicrobial exposure, this is a practical approach to inform future research and stewardship.

Data Availability

All data produced in the present study are available upon reasonable request to the authors

Acknowledgements

We would like to thank all the participants who took part in ARMORD, and also the nurses, healthcare support workers, and doctors on the OUH Haematology ward for their help with this study. This work was supported by the National Institute for Health Research Health Protection Research Unit (NIHR HPRU) in Healthcare Associated Infections and Antimicrobial Resistance at Oxford University in partnership with the UK Health Security Agency (NIHR200915), and the NIHR Biomedical Research Centre, Oxford. ASW is an NIHR Senior Investigator. The views expressed are those of the authors and not necessarily those of the NHS, the NIHR, the Department of Health or the UK Health Security Agency. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Declaration of interests

No authors have no conflict of interest to declare.

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Reviewer #1 (Public Review):

Summary:

In this manuscript, the authors provide a study among healthy individuals, general medical patients and patients receiving haematopoietic cell transplants (HCT) to study the gut microbiome through shotgun metagenomic sequencing of stool samples. The first two groups were sampled once, while the patients receiving HCT were sampled longitudinally. A range of metadata (including current and previous (up to 1 year before sampling) antibiotic use) was recorded for all sampled individuals. The authors then performed shotgun metagenomic sequencing (using the Illumina platform) and performed bioinformatic analyses on these data to determine the composition and diversity of the gut microbiota and the antibiotic resistance genes therein. The authors conclude, on the basis of these analyses, that some antibiotics had a large impact on gut microbiota diversity, and could select opportunistic pathogens and/or antibiotic resistance genes in the gut microbiota.

Strengths:

The major strength of this study is the considerable achievement of performing this observational study in a large cohort of individuals. Studies into the impact of antibiotic

therapy on the gut microbiota are difficult to organise, perform and interpret, and this work follows state-of-the-art methodologies to achieve its goals. The authors have achieved their objectives and the conclusion they draw on the impact of different antibiotics and their impact on the gut microbiota and its antibiotic resistance genes (the 'resistome', in short), are supported by the data presented in this work.

Weaknesses:

The weaknesses are the lack of information on the different resistance genes that have been identified and which could have been supplied as Supplementary Data. In addition, no attempt is made to assess whether the identified resistance genes are associated with mobile genetic elements and/or (opportunistic) pathogens in the gut. While this is challenging with short-read data, alternative approaches like long-read metagenomics, Hi-C and/or culture-based profiling of bacterial communities could have been employed to further strengthen this work. Unfortunately, the authors have not attempted to perform corrections for multiple testing because many antibiotic exposures were correlated.

Impact:

The work may impact policies on the use of antibiotics, as those drugs that have major impacts on the diversity of the gut microbiota and select for antibiotic resistance genes in the gut are better avoided. However, the primary rationale for antibiotic therapy will remain the clinical effectiveness of antimicrobial drugs, and the impact on the gut microbiota and resistome will be secondary to these considerations.

<https://doi.org/10.7554/eLife.97751.1.sa1>

Reviewer #2 (Public Review):

Summary:

In this manuscript by Peto et al., the authors describe the impact of different antimicrobials on gut microbiota in a prospective observational study of 225 participants (healthy volunteers, inpatients and outpatients). Both cross-sectional data (all participants) and longitudinal data (a subset of 79 haematopoietic cell transplant patients) were used. Using metagenomic sequencing, they estimated the impact of antibiotic exposure on gut microbiota composition and resistance genes. In their models, the authors aim to correct for potential confounders (e.g. demographics, non-antimicrobial exposures and physiological abnormalities), and for differences in the recency and total duration of antibiotic exposure. I consider these comprehensive models an important strength of this observational study. Yet, the underlying assumptions of such models may have impacted the study findings (detailed below). Other strengths include the presence of both cross-sectional and longitudinal exposure data and the presence of both healthy volunteers and patients. Together, these observational findings expand on previous studies (both observational and RCTs) describing the impact of antimicrobials on gut microbiota.

Weaknesses:

(1) The main weaknesses result from the observational design. This hampers causal interpretation and corrects for potential confounding necessary. The authors have used comprehensive models to correct for potential confounders and for differences between participants in duration of antibiotic exposure and time between exposure and sample collection. I wonder if some of the choices made by the authors did affect these findings. For example, the authors did not include travel in the final model, but travel (most importantly, south Asia) may result in the acquisition of AMR genes [Worby et al., *Lancet Microbe* 2023; PMID 37716364]. Moreover, non-antimicrobial drugs (such as proton pump inhibitors) were

not included but these have a well-known impact on gut microbiota and might be linked with exposure to antimicrobial drugs. Residual confounding may underlie some of the unexplained discrepancies between the cross-sectional and longitudinal data (e.g. for vancomycin).

In addition, the authors found a disruption half-life of 6 days to be the best fit based on Shannon diversity. If I'm understanding correctly, this results in a near-zero modelled exposure of a 14-day-course after 70 days (purple line; Supplementary Figure 2). However, it has been described that microbiota composition and resistome (not Shannon diversity!) remain altered for longer periods of time after (certain) antibiotic exposures (e.g. Anthony et al., Cell Reports 2022; PMID 35417701). The authors did not assess whether extending the disruption half-life would alter their conclusions.

(2) Another consequence of the observational design of this study is the relatively small number of participants available for some comparisons (e.g. oral clindamycin was only used by 6 participants). Care should be taken when drawing any conclusions from such small numbers.

(3) The authors assessed log-transformed relative abundances of specific bacteria after subsampling to 3.5 million reads. While I agree that some kind of data transformation is probably preferable, these methods do not address the compositional data of microbiome data and using a pseudocount (10⁻⁶) is necessary for absent (i.e. undetected) taxa [Gloor et al., Front Microbiol 2017; PMID 29187837]. Given the centrality of these relative abundances to their conclusions, a sensitivity analysis using compositionally-aware methods (such as a centred log-ratio (clr) transformation) would have added robustness to their findings.

(4) An overall description of gut microbiota composition and resistome of the included participants is missing. This makes it difficult to compare the current study population to other studies. In addition, for correct interpretation of the findings, it would have been helpful if the reasons for hospital visits of the general medical patients were provided.

<https://doi.org/10.7554/eLife.97751.1.sa0>